

5. Project Development Phase

FITBUDDY – AI FITNESS PLAN GENERATOR USING GEMINI MODELS

- **Backend Development:** Implement the FastAPI application, routes, request handling, templates, database connection, and application services.
- **AI Integration:** Connect the application to Google Gemini through a server-side API configuration. Prompts should request structured and useful fitness-plan content based on the user's supplied information.
- **Frontend Development:** Create HTML/Jinja2 pages and CSS for user input, plan display, feedback, and administrator views.
- **Database Development:** Define SQLAlchemy models and CRUD operations for storing users and generated plans.
- **Feedback Feature:** Allow the user to request changes such as more cardio or additional rest days, then generate an updated plan using the feedback.
- **Configuration:** Keep API keys and other secrets in environment variables rather than hard-coding them in source files.